



Software Development Proposal

Deloitte.

1. Overview

The IIoT (industrial internet of things) are a specialised subset of IoT (internet of things) that primarily focus on operational efficiency, predictive maintenance and automation in industrial settings. Daikibo's four factories, those being, factory Meiyo, Seiko, Berlin and Shenzhen utilise the following nine machines; CNC, laser cutter, heavy duty drill spot welder, laser welder, metal press, furnace, conveyor belt and the air wrench. Failures of these machines lead to severe impacts on the operations of Daikibo's factories, such as halted assembly lines, inventory losses and even hazardous working conditions. Therefore, the requirement of a dashboard to constantly monitor the health status of these equipment is vital.

It is also crucial that the dashboard is secure. Hence, access to the page should only happen within Daikibo's factories' intranet and authentication should be synced to an internal authentication server.

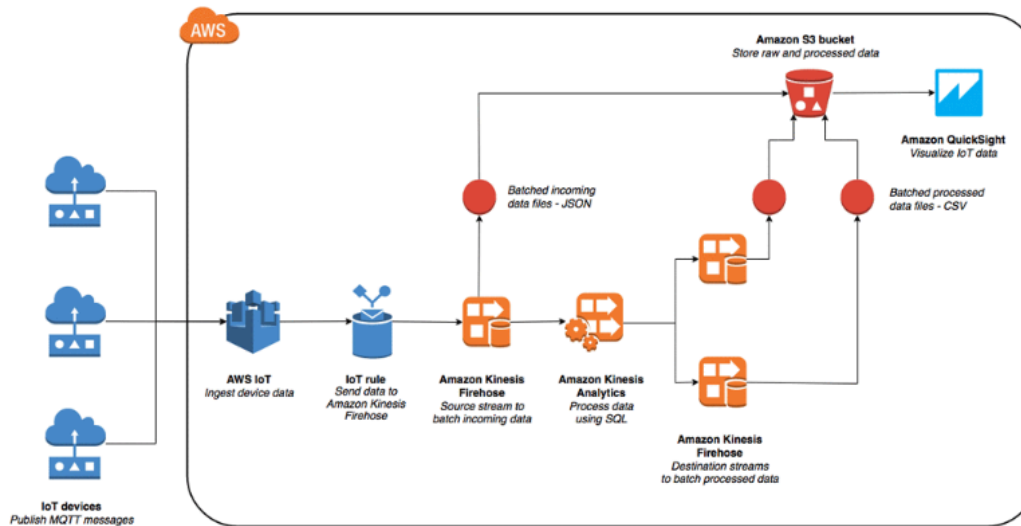
Some other requirements of this dashboard are that the dashboard consists of a single page listing the current status of all monitored devices, the view being able to be collapsed/expanded at the factory and device levels as well as showing the history of statuses.

The solution that I propose to Daikibo is to implement the AWS (Amazon Web Services) IoT Core product. This product satisfies the key requirements of Daikibo, in addition, it goes above and beyond the requirements, providing numerous benefits which I will discuss in the forthcoming sections.

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Key Requirements Addressed:

Private dashboard showing the health status of the 9 machines in each of Daikibo's 4 factories for which they collect telemetry - using AWS serverless and managed services, you can easily build a dashboard



(Desai, 2017)

Access to the page happens only within the client's intranet – although AWS is a cloud platform, the dashboard will not be exposed to the public internet. A private connection between Daikibo's factory networks and AWS will be established using AWS Site-to-Site VPN and AWS PrivateLink, meaning the dashboard is only reachable from within Daikibo's intranet.

Authentication is synced to an internal authentication server – AWS IAM Identity Center supports federation via SAML 2.0, allowing it to be synced with Daikibo's internal authentication server. Employees can therefore sign in using their existing company-wide accounts, with no additional credentials to manage.

The dashboard consists of a single page listing the current status of all monitored devices – a lightweight single-page web application will display the live status of all thirty-six machines across the four factories.

The view is collapsible/expandable at the factory and device levels – telemetry ingested through AWS IoT Core will be stored, enabling each device to be expanded to reveal a full history of its statuses.

2. Scope

The scope of this project encompasses the design, development, testing and integration of a private, single-page health-monitoring dashboard for the thirty-six machines across Daikibo's four factories. Telemetry from each machine will be ingested via AWS IoT Core using the MQTT protocol, processed through Amazon Kinesis (Desai, 2017), and stored to preserve a complete history of device statuses.

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As illustrated in the wireframe on the following page, the dashboard presents all four factories on a single page. Each factory can be collapsed or expanded to reveal its nine machines, and each machine can, in turn, be expanded to display a chronological history of its health statuses. A machine's current status is indicated at a glance, allowing operators to identify failures or degradation immediately. The page will be accessible exclusively within Daikibo's intranet, and users will authenticate using their existing company-wide accounts.

Out of scope for this phase are automated alerting, predictive maintenance analytics and mobile applications, though the proposed architecture readily supports these as future enhancements.

The proposed solution is also proven at an industrial scale. The Volkswagen Group, Europe's largest car maker, built its Industrial Cloud on AWS, utilising AWS IoT services to connect data from machines across more than 120 factory sites, targeting a 30 percent increase in productivity and a 30 percent decrease in factory costs (Amazon Web Services, 2026). Similarly, Traeger Grills migrated over 110,000 devices to AWS IoT Core in just three months with zero downtime, whilst cutting costs per device by 90 percent (Amazon Web Services, 2020). These cases demonstrate that the solution I propose is reliable, cost-effective and capable of scaling far beyond Daikibo's current requirements.

✓  Daikibo Factory Meiyo	Last update: <1min ago ◀
✓  Daikibo Factory Seiko	Last update: <1min ago ◀
✓  Daikibo Berlin	Last update: <1min ago ◀
✗  Daikibo Shenzhen	Last update: <1min ago ▼
✗  CNC	Last update: 2min ago ▼
✗  Status: Unhealthy	2min ago
✓  Status: Healthy	12min ago
Load More	
✓  LaserCutter	Last update: <1min ago ◀
✓  HeavyDutyDrill	Last update: <1min ago ◀
✓  SpotWelder	Last update: <1min ago ◀
✓  LaserWelder	Last update: <1min ago ◀
✓  MetalPress	Last update: <1min ago ◀
✓  Furnace	Last update: <1min ago ◀
✓  ConveyorBelt	Last update: <1min ago ◀
✓  AirWrench	Last update: <1min ago ◀

3. Estimate

I estimate the project will require a total of 620 man-hours, broken down as follows: development of the dashboard and the AWS telemetry pipeline will require 380 hours; testing, comprising unit, system and user-acceptance testing, will require 140 hours; and integration of the product into Daikibo's intranet, including the VPN configuration and synchronisation with the internal authentication server, will require 100 hours.

4. Timeline

1. [1st of September 2026] Design starts
2. [15th of September 2026] Design sign-off; development starts
3. [3rd of November 2026] Development complete; testing starts
4. [24th of November 2026] Testing complete; integration starts
5. [15th of December 2026] Integration into Daikibo's intranet complete; user-acceptance testing
6. [22nd of December 2026] Go-live

5. Support

Following go-live, Daikibo can rely on continuous product support. This includes the prompt resolution of bug fixes, a dedicated support-ticket channel with guaranteed response times, and regular maintenance of the underlying AWS infrastructure. Furthermore, as Daikibo's operational needs evolve, new functionality, such as automated alerting or predictive maintenance, can be scoped and delivered as incremental enhancements to the existing platform.

6. References

Desai, K. (2017) Build a visualization and monitoring dashboard for IoT data with Amazon Kinesis Analytics and Amazon QuickSight, AWS Big Data Blog, 17 May. Available at: <https://aws.amazon.com/blogs/big-data/build-a-visualization-and-monitoring-dashboard-for-iot-data-with-amazon-kinesis-analytics-and-amazon-quicksight/> (Accessed: 2 July 2026).

Amazon Web Services (2020) *OST helps Traeger Grills migrate over 110K devices to AWS IoT with zero downtime*. Available at: <https://aws.amazon.com/partners/success/traeger-grills-ost/> (Accessed: 3 July 2026).

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Amazon Web Services (2026) *The Volkswagen Group on AWS*. Available at:

<https://aws.amazon.com/solutions/case-studies/innovators/volkswagen-group/> (Accessed: 3 July 2026).